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## DSC 40B - Quiz 01

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Name:

Student ID:

**About the quizzes:**

- Quizzes in DSC 40B are *optional* and graded pass/fail.
- A score of 70% or higher earns a “pass” and 1.5 credits toward your final grade.
- If you don’t pass, no credits are earned, but it doesn’t hurt your grade.
- You have 30 minutes to complete the quiz.
- Unfortunately, we can’t answer clarifying questions during the quiz. If you think a question has a bug or is unclear, please let us know in a private post on Campuswire after the quiz, and we’ll take it into account when grading.

**Problem 1.**

What is the time complexity of the following function?

```
def foo(arr):  
    """`arr` is a list containing n numbers."""  
    x = sum(arr) - min(arr)  
    if max(arr) > 100:  
        return x  
    else:  
        return -x
```

- ☐  $\Theta(1)$     ☐  $\Theta(\log n)$     ☐  $\Theta(n)$     ☐  $\Theta(n \log n)$     ☐  $\Theta(n^2)$     ☐  $\Theta(n^3)$     ☐  $\Theta(n^4)$

**Problem 2.**

What is the time complexity of the following function?

```
def foo(arr):  
    """`arr` is a list containing n numbers."""  
    for x in arr:  
        if x > sum(arr) / 2:  
            print('big!')
```

- ☐  $\Theta(1)$     ☐  $\Theta(\log n)$     ☐  $\Theta(n)$     ☐  $\Theta(n \log n)$     ☐  $\Theta(n^2)$     ☐  $\Theta(n^3)$     ☐  $\Theta(n^4)$

**Problem 3.**

What is the time complexity of the following function?

```
def foo(n):  
    for i in range(n**2):  
        for j in range(n):  
            for k in range(5):  
                print(i + j + k)
```

- ☐  $\Theta(1)$     ☐  $\Theta(\log n)$     ☐  $\Theta(n)$     ☐  $\Theta(n \log n)$     ☐  $\Theta(n^2)$     ☐  $\Theta(n^3)$     ☐  $\Theta(n^4)$

**Problem 4.**

What is the time complexity of the following function? (Recall that `//` is integer division: `n // 100` divides `n` by 100 and rounds down to the nearest integer.)

```
def foo(n):  
    for i in range(n**2 - 3*n + 50):  
        for j in range(n // 100):  
            print(i + j)
```

- ☐  $\Theta(1)$     ☐  $\Theta(\log n)$     ☐  $\Theta(n)$     ☐  $\Theta(n \log n)$     ☐  $\Theta(n^2)$     ☐  $\Theta(n^3)$     ☐  $\Theta(n^4)$

**Problem 5.**

What is the time complexity of the following function?

```
def foo(n):  
    for i in range(n):  
        for j in range(n):  
            print(i + j)  
        for k in range(n**2):  
            print(i + k)
```

☐  $\Theta(1)$     ☐  $\Theta(\log n)$     ☐  $\Theta(n)$     ☐  $\Theta(n \log n)$     ☐  $\Theta(n^2)$     ☐  $\Theta(n^3)$     ☐  $\Theta(n^4)$

**Problem 6.**

What is the time complexity of the following function?

```
def foo(n):  
    i = n  
    while i > 1:  
        i = i / 3  
        print(i)
```

☐  $\Theta(1)$     ☐  $\Theta(\log n)$     ☐  $\Theta(n)$     ☐  $\Theta(n \log n)$     ☐  $\Theta(n^2)$     ☐  $\Theta(n^3)$     ☐  $\Theta(n^4)$

**Problem 7.**

What is the time complexity of the following function?

```
def foo(n):  
    i = 1  
    while i < n:  
        i = i * 3  
        for j in range(n):  
            print(i + j)
```

☐  $\Theta(1)$     ☐  $\Theta(\log n)$     ☐  $\Theta(n)$     ☐  $\Theta(n \log n)$     ☐  $\Theta(n^2)$     ☐  $\Theta(n^3)$     ☐  $\Theta(n^4)$

**Problem 8.**

What is the time complexity of the following function?

```
def foo(n):  
    for i in range(n):  
        for j in range(i, n):  
            for k in range(10):  
                print(i + j + k)
```

☐  $\Theta(1)$     ☐  $\Theta(\log n)$     ☐  $\Theta(n)$     ☐  $\Theta(n \log n)$     ☐  $\Theta(n^2)$     ☐  $\Theta(n^3)$     ☐  $\Theta(n^4)$

**Problem 9.**

What is the time complexity of the following function?

```
def foo(n):  
    for i in range(n**2):  
        for j in range(i):  
            print(i + j)
```

- ☐  $\Theta(1)$     ☐  $\Theta(\log n)$     ☐  $\Theta(n)$     ☐  $\Theta(n \log n)$     ☐  $\Theta(n^2)$     ☐  $\Theta(n^3)$     ☐  $\Theta(n^4)$

**Problem 10.**

*Exactly* how many times is `print` called when `foo(n)` is run?

```
def foo(n):  
    for i in range(n):  
        for j in range(i + 1, n):  
            print(i, j)
```

- ☐  $n^2$     ☐  $\frac{n(n-1)}{2}$     ☐  $\frac{n(n+1)}{2}$     ☐  $\frac{n^2}{4}$     ☐  $n(n-1)$